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course: BBT.MJS.101 SMALL SAMPLES DATA ANALYSIS
Software used: SPSS

PROJECT WORK

1. Descriptive statistics

1.1)

CAD		Mean		95% Confidence Interval		Std. Deviation
		Statistic	Std. Error	Lower	Upper	
Age	0	48.38	1.214	45.97	50.79	12.138
	1	53.98	0.839	52.31	55.65	8.394
RWA	0	0.1175	0.01471	0.0883	0.1467	0.14714
	1	0.1920	0.01458	0.1631	0.2209	0.14580
ST/HR	0	0.9105	0.08597	0.7399	1.0811	0.85966
	1	3.2413	0.25144	2.7424	3.7402	2.51438
ST-DEP	0	0.0537	0.00616	0.0415	0.0659	0.06164
	1	0.1369	0.01130	0.1145	0.1593	0.11304
MaxHR	0	159.07	2.120	154.86	163.28	21.204
	1	122.74	2.168	118.44	127.04	21.676

1: CAD disease

0: no CAD disease

Descriptive statistics were calculated for age, RWA, ST/HR, ST depression, and maxHR according to CAD status. The results show that patients with CAD were older on average than patients without CAD. The mean age in the CAD group was 53.98 years (SD = 8.39, 95% CI: 52.31–55.65), whereas the mean age in the no-CAD group was 48.38 years (SD = 12.14, 95% CI: 45.97–50.79).

For RWA, the mean value was higher in patients with CAD (mean = 0.1920, SD = 0.1458, 95% CI: 0.1631–0.2209) than in patients without CAD (mean = 0.1175, SD = 0.1471, 95% CI: 0.0883–0.1467). The standard deviations were relatively larger in the two groups, and they may overlap means of each group.

A similar pattern was seen for the ST/HR index. Patients with CAD had a much higher mean ST/HR value (mean = 3.2413, SD = 2.5144, 95% CI: 2.7424–3.7402) than patients without CAD (mean = 0.9105, SD = 0.8597, 95% CI: 0.7399–1.0811). The variability was also clearly greater in the CAD group.

For ST depression, patients with CAD again showed higher values. The mean ST depression was 0.1369 (SD = 0.1130, 95% CI: 0.1145–0.1593) in the CAD group and 0.0537 (SD = 0.0616, 95% CI: 0.0415–0.0659) in the no-CAD group.

Patients without CAD had the higher maxHR. The mean maxHR was 159.07 (SD = 21.20, 95% CI: 154.86–163.28) in the no-CAD group, compared with 122.74 (SD = 21.68, 95% CI: 118.44–127.04) in the CAD group. The standard deviations were similar in both groups, but the means were far apart.

Overall, the descriptive statistics suggest that compared with patients without CAD, patients with CAD were older, had higher RWA, higher ST/HR index, and higher ST depression, while patients without CAD had a markedly higher maximum heart rate

MI * Sex Crosstabulation

Count		Sex		Total
		F	M	
MI	0	69	88	157
	1	4	16	20
	2	1	22	23
Total		74	126	200

MI * CAD Crosstabulation

Count		CAD		Total
		0	1	
MI	0	100	57	157
	1	0	20	20
	2	0	23	23
Total		100	100	200

Digit * Sex Crosstabulation

Count		Sex		Total
		F	M	
Digit	0	73	124	197
	1	1	2	3
Total		74	126	200

Digit * CAD Crosstabulation

Count		CAD		Total
		0	1	
Digit	0	99	98	197
	1	1	2	3
Total		100	100	200

Nitro * Sex Crosstabulation

Count		Sex		Total
		F	M	
Nitro	0	52	71	123
	1	22	55	77
Total		74	126	200

Nitro * CAD Crosstabulation

Count		CAD		Total
		0	1	
Nitro	0	97	26	123
	1	3	74	77
Total		100	100	200

Calsi * Sex Crosstabulation

Count		Sex		Total
		F	M	
Calsi	0	61	102	163
	1	13	24	37
Total		74	126	200

Calsi * CAD Crosstabulation

Count		CAD		Total
		0	1	
Calsi	0	98	65	163
	1	2	35	37
Total		100	100	200

Beta * Sex Crosstabulation

Count		Sex		Total
		F	M	
Beta	0	51	62	113
	1	23	64	87
Total		74	126	200

Beta * CAD Crosstabulation

Count		CAD		Total
		0	1	
Beta	0	95	18	113
	1	5	82	87
Total		100	100	200

The categorical descriptive statistics show clear differences between the CAD and no-CAD groups. myocardial infarction (MI) was observed only in patients with CAD. All patients without CAD had no infarction, while among CAD patients 57 had no infarction, 20 had recent infarction, and 23 had old infarction. This suggests that myocardial infarction was much more common in the CAD group. And more than double of CAD patients had no previous record of MI. Further, More males had myocardial infarction symptoms than females.

For medication use, digitalis was very rare in both groups, with only 1 no-CAD patient and 2 CAD patients using it. Nitro use was much more common in CAD patients, with 74 CAD patients using nitro compared with only 3 no-CAD patients. Calcium antagonist use also showed the same pattern, with 35 CAD patients using it compared with only 2 no-CAD patients. Beta blocker use was also much higher in the CAD group, with 82 CAD patients using beta blockers compared with 5 no-CAD patients.

In terms of sex distribution, males were more common than females in the total sample. Males also appeared more frequently among patients with myocardial infarction and among users of most medications. Overall, the categorical data suggest that MI history and the use of nitro, calcium antagonists, and beta blockers were more common in patients with CAD, while digitalis use was uncommon in both groups.

1.2)

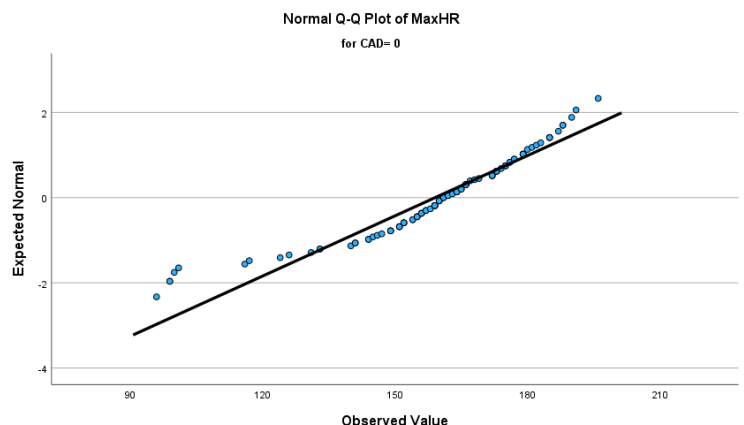
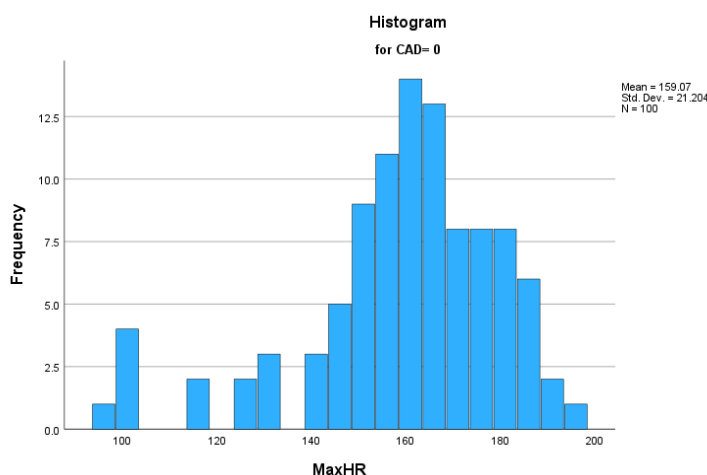
The normality of MaxHR was tested separately in patients without CAD and in patients with CAD using the Shapiro–Wilk test.

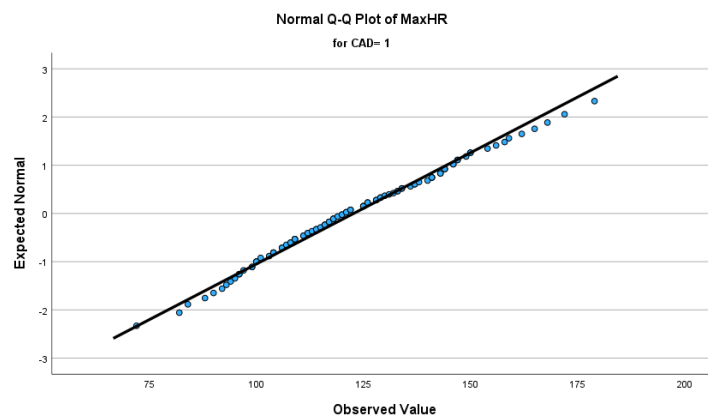
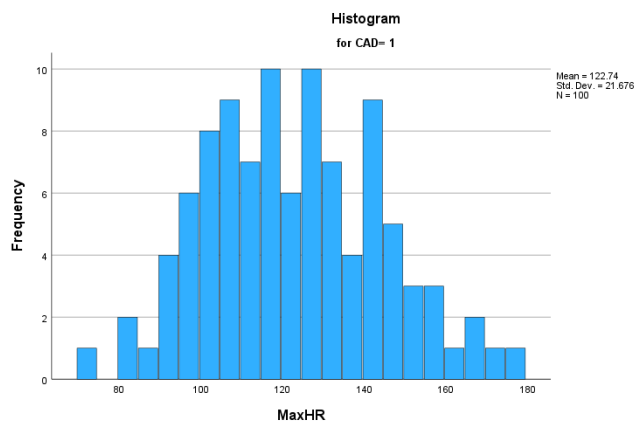
Tests of Normality							
		Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	CAD	Statistic	df	Sig.	Statistic	df	Sig.
MaxHR	0	0.122	100	<.001	0.915	100	<.001
	1	0.057	100	0.200*	0.991	100	0.737

1: CAD disease;
0: no CAD disease

*. This is a lower bound of the true significance.

a. Lilliefors Significance Correction





H0: MaxHR is normally distributed.

H1: MaxHR is not normally distributed.

The normality of MaxHR was tested separately in patients without CAD and in patients with CAD using the Shapiro–Wilk test and the Kolmogorov–Smirnov test.

For the non-CAD group, both tests gave p-values less than 0.05. The Shapiro–Wilk test was significant ($W = 0.915$, $p < 0.001$) and the Kolmogorov–Smirnov test was also significant ($p < 0.001$). Therefore, the null hypothesis of normality is rejected, and MaxHR in patients without CAD is not normally distributed. The histogram and Q-Q plot also support this result.

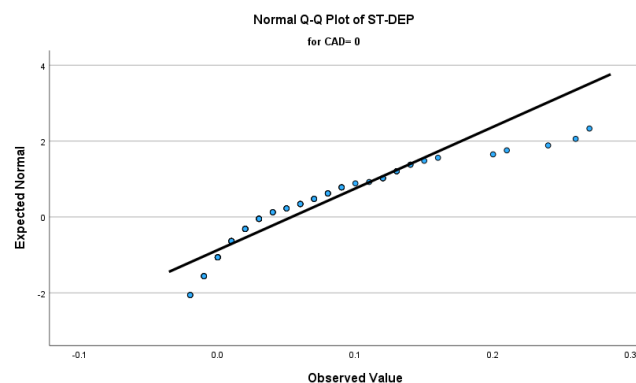
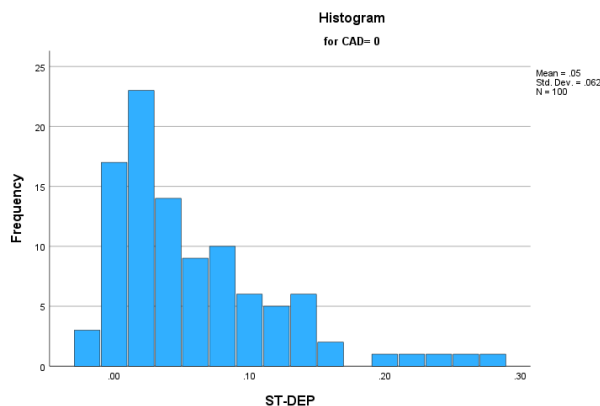
For the CAD group, the Shapiro–Wilk test was not significant ($W = 0.991$, $p = 0.737$), which suggests that MaxHR in this group is approximately normally distributed. The Kolmogorov–Smirnov test gave $p = 0.200$, which is also greater than 0.05, so this test also does not reject normality. Therefore, in the CAD group, MaxHR can be considered approximately normal. The histogram also looks fairly close to a symmetric distribution.

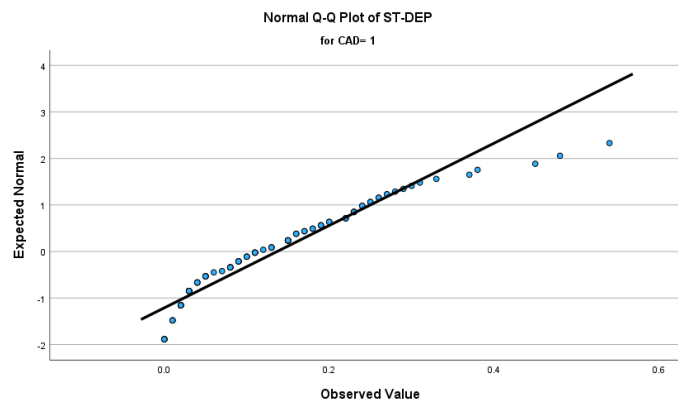
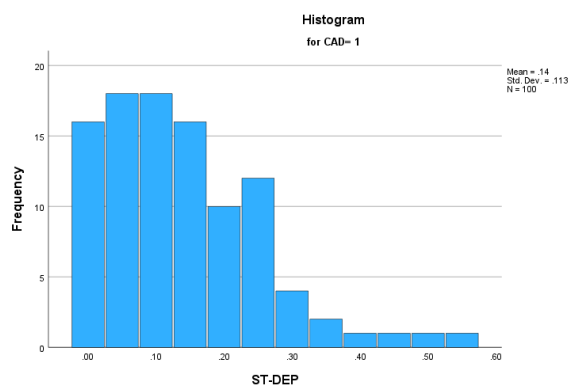
Overall, MaxHR is not normal in the non-CAD group but is approximately normal in the CAD group. Since the normality assumption is not satisfied in both groups, it would be safer to use a non-parametric test.

2. Comparing the distributions of two groups

2.1)

The normality of ST depression was tested separately in patients without CAD and in patients with CAD using the Shapiro–Wilk test.





Tests of Normality

		Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	CAD	Statistic	df	Sig.	Statistic	df	Sig.
ST-DEP	0	.180	100	<.001	.860	100	<.001
	1	.113	100	.003	.909	100	<.001

a. Lilliefors Significance Correction

ST depression

H0: ST depression is normally distributed.

H1: ST depression is not normally distributed.

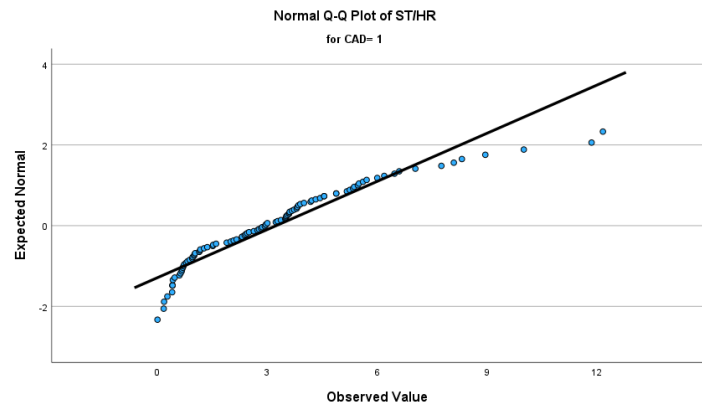
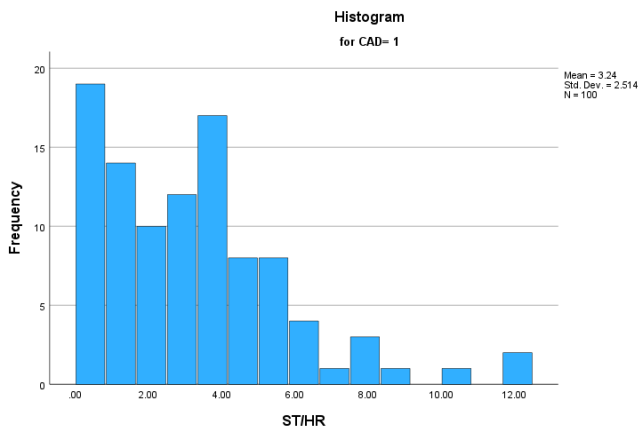
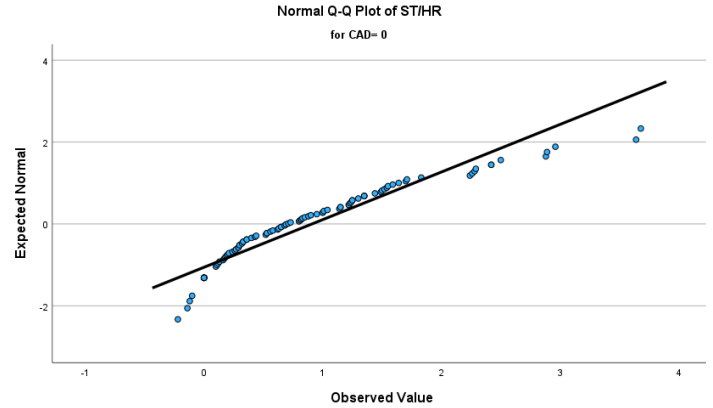
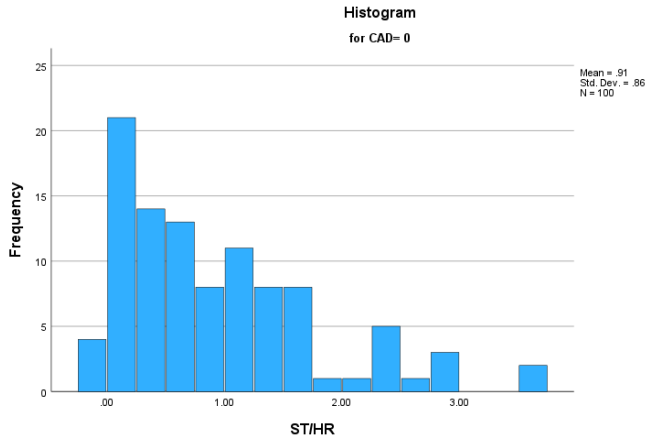
The Shapiro–Wilk test showed p-values less than 0.05 in both CAD and non-CAD groups. Therefore, the null hypothesis was rejected, and ST depression was considered **not normally distributed** in both groups.

The normality of ST/HR index and RWA was tested separately in patients without CAD and in patients with CAD using the Shapiro–Wilk test.

Tests of Normality

		Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	CAD	Statistic	df	Sig.	Statistic	df	Sig.
ST/HR	0	.105	100	.009	.901	100	<.001
	1	.107	100	.007	.902	100	<.001
RWA	0	.177	100	<.001	.772	100	<.001
	1	.111	100	.004	.956	100	.002

a. Lilliefors Significance Correction

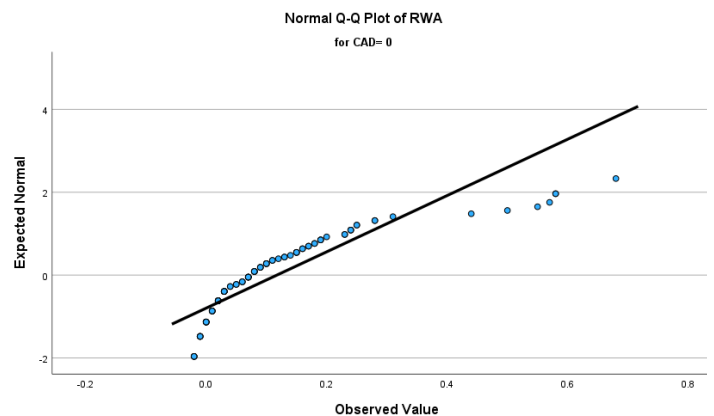
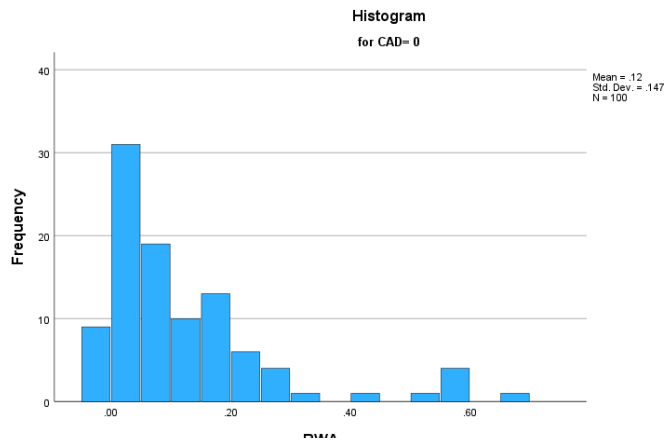


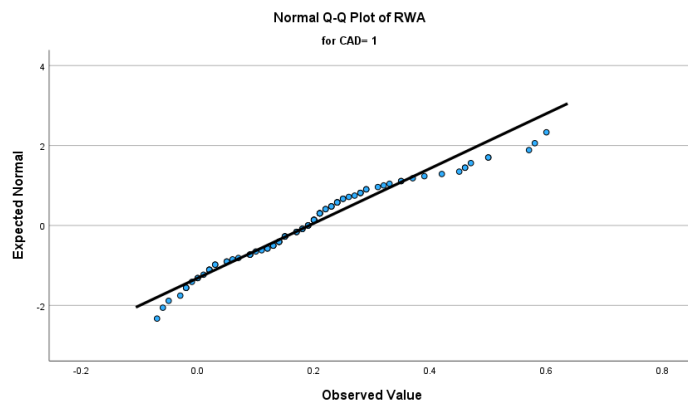
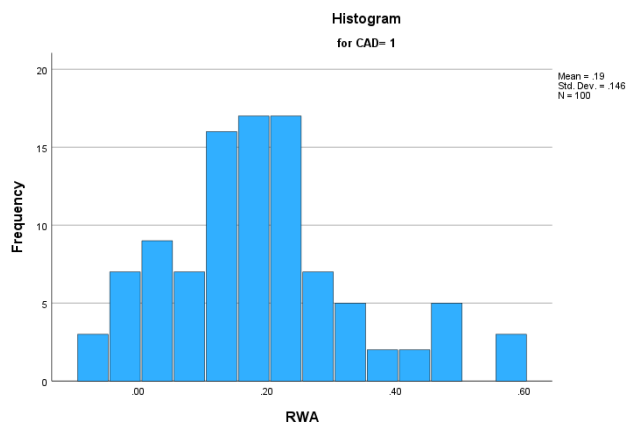
ST/HR index:

H0: ST/HR index is normally distributed.

H1: ST/HR index is not normally distributed.

The Shapiro–Wilk test gave p-values less than 0.05 for both groups. This means the null hypothesis of normality was rejected. So, the ST/HR index was not normally distributed in CAD and non-CAD patients.





RWA:

H0: RWA is normally distributed.

H1: RWA is not normally distributed.

The Shapiro–Wilk test also showed p-values less than 0.05 in both groups. Therefore, the null hypothesis was rejected, and RWA was considered not normally distributed in both CAD and non-CAD groups.

The Mann–Whitney U test was used to compare patients with CAD and patients without CAD in terms of the distributions of ST depression, ST/HR index, Δ RWA, and maxHR, because these variables were not normally distributed. The test is appropriate because the CAD and non-CAD groups are independent.

Mann-Whitney Test

		Ranks		
	CAD	N	Mean Rank	Sum of Ranks
MaxHR	0	100	138.62	13862.00
	1	100	62.38	6238.00
	Total	200		

Test Statistics

	MaxHR
Mann-Whitney U	1188.000
Wilcoxon W	6238.000
Z	-9.315
Asymp. Sig. (2-tailed)	<.001

a. Grouping Variable: CAD

H0: Distribution is the same in CAD and no-CAD groups.

H1: Distribution is different in CAD and no-CAD groups.

The test showed a statistically significant difference between the two groups ($U = 1188.000$, $Z = -9.315$, $p < 0.001$). Since $p < 0.05$, the null hypothesis was rejected. This indicates that MaxHR differed significantly between CAD and no-CAD patients. The mean rank of MaxHR was higher in patients without CAD (138.62) than in patients with CAD (62.38), showing that maximum heart rate values were generally higher in patients without CAD.

Mann-Whitney Test

	CAD	N	Mean Rank	Sum of Ranks
ST-DEP	0	100	75.83	7582.50
	1	100	125.18	12517.50
	Total	200		
ST/HR	0	100	68.09	6808.50
	1	100	132.92	13291.50
	Total	200		
RWA	0	100	82.18	8217.50
	1	100	118.83	11882.50
	Total	200		

Test Statistics

	ST-DEP	ST/HR	RWA
Mann-Whitney U	2532.500	1758.500	3167.500
Wilcoxon W	7582.500	6808.500	8217.500
Z	-6.039	-7.921	-4.480
Asymp. Sig. (2-tailed)	<.001	<.001	<.001

a. Grouping Variable: CAD

H0: Distribution is the same in CAD and no-CAD groups. **H1:** Distribution is different in CAD and no-CAD groups.

ST depression: The test showed a statistically significant difference between the two groups ($U = 2532.500$, $Z = -6.039$, $p < 0.001$). Since the p-value was less than 0.05, the null hypothesis was rejected. This means that the distribution of ST depression was not the same in the CAD and no-CAD groups. The mean rank of ST depression was higher in patients with CAD (125.18) than in patients without CAD (75.83), indicating that ST depression values were generally higher in patients with CAD.

ST/HR index: The test showed a statistically significant difference between the two groups ($U = 1758.500$, $Z = -7.921$, $p < 0.001$). Since the p-value was less than 0.05, the null hypothesis was rejected. This means that the distribution of the ST/HR index was not the same in the CAD and no-CAD groups. The mean rank of the ST/HR index was higher in patients with CAD (132.92) than in patients without CAD (68.09), indicating that ST/HR index values were generally higher in patients with CAD.

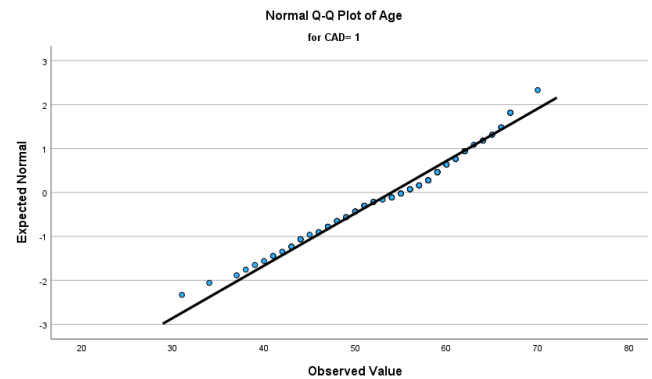
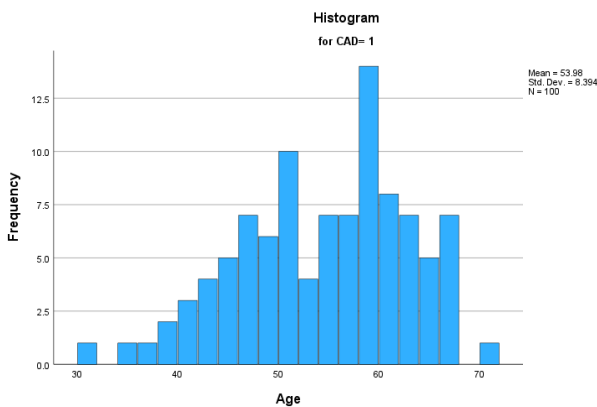
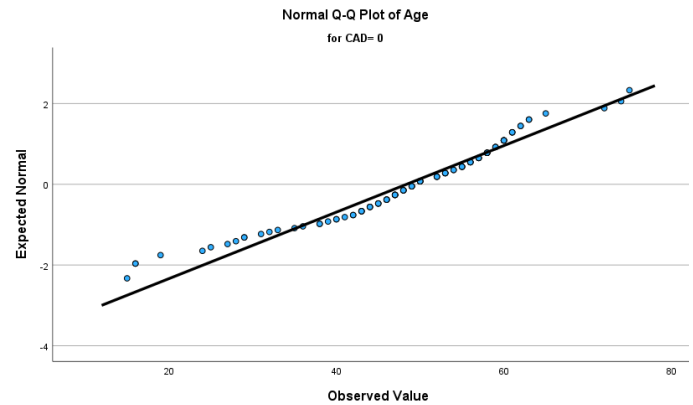
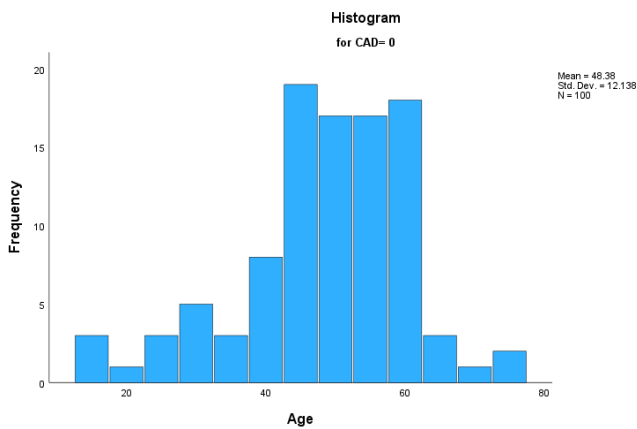
ΔRWA: The test showed a statistically significant difference between the two groups ($U = 3167.500$, $Z = -4.480$, $p < 0.001$). Since the p-value was less than 0.05, the null hypothesis was rejected. This means that the distribution of ΔRWA was not the same in the CAD and no-CAD groups. The mean rank of ΔRWA was higher in patients with CAD (118.83) than in patients without CAD (82.18), indicating that ΔRWA values were generally higher in patients with CAD.

2.2)

Tests of Normality

	CAD	Kolmogorov-Smirnov ^a			Shapiro-Wilk		
		Statistic	df	Sig.	Statistic	df	Sig.
Age	0	.099	100	.018	.955	100	.002
	1	.104	100	.010	.976	100	.061

a. Lilliefors Significance Correction



H0: Age is normally distributed.

H1: Age is not normally distributed.

A normality test was performed for age in the CAD and no-CAD groups. In the no-CAD group, the Shapiro–Wilk p-value was less than 0.05, Reject H0, which indicates that age was not normally distributed in this group. In the CAD group, the Shapiro–Wilk p-value was greater than 0.05, cannot reject H0, which suggests that age was approximately normally distributed. However, because normality was not satisfied in both groups, it was safer to use a non-parametric test. Since the CAD and no-CAD groups are independent, the Mann–Whitney U test was used to compare age between the two groups

Mann-Whitney Test for Age

Ranks				
	CAD	N	Mean Rank	Sum of Ranks
Age	0	100	86.40	8640.00
	1	100	114.60	11460.00
	Total	200		

Test Statistics	
	Age
Mann-Whitney U	3590.000
Wilcoxon W	8640.000
Z	-3.447
Asymp. Sig. (2-tailed)	<.001

a. Grouping Variable: CAD

H0: Distribution is the same in CAD and no-CAD groups. **H1:** Distribution is different in CAD and no-CAD groups

The test showed a statistically significant difference between the two groups ($U = 3590.000$, $Z = -3.447$, $p < 0.001$). Since the p-value was less than 0.05, the null hypothesis was rejected. This means that the age distribution was not the same in the CAD and no-CAD groups. The mean rank of age was higher in patients with CAD (114.60) than in patients without CAD (86.40), indicating that CAD patients were generally older than no-CAD patients

A chi-square test was used to compare sex distribution between CAD and no-CAD patients.

CAD * Sex Crosstabulation

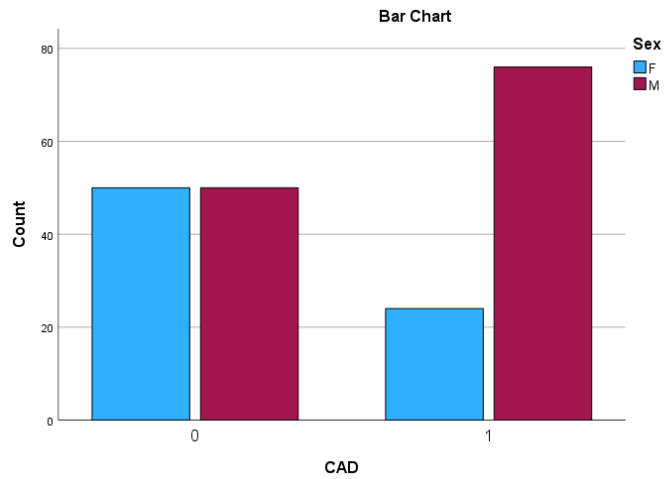
			Sex		Total
			F	M	
CAD	0	Count	50	50	100
		% within CAD	50.0%	50.0%	100.0%
	1	Count	24	76	100
		% within CAD	24.0%	76.0%	100.0%
Total	Count		74	126	200
	% within CAD		37.0%	63.0%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2- sided)	Exact Sig. (2- sided)	Exact Sig. (1- sided)
Pearson Chi-Square	14.500 ^a	1	<.001		
Continuity Correction ^b	13.406	1	<.001		
Likelihood Ratio	14.737	1	<.001		
Fisher's Exact Test				<.001	<.001
N of Valid Cases	200				

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 37.00.

b. Computed only for a 2x2 table



H0: Sex distribution is the same in CAD and no-CAD groups.

H1: Sex distribution is different in CAD and no-CAD groups.

A chi-square test of independence was used to compare sex distribution between CAD and no-CAD patients. The test results shows a statistically significant association between sex and CAD status, $\chi^2(1) = 14.50$, $p < 0.001$. Since the p-value was less than 0.05, the null hypothesis was rejected. This means that sex distribution was not the same in the CAD and no-CAD groups. In the no-CAD group, females and males were equally distributed, with 50 females (50.0%) and 50 males (50.0%). In the CAD group, there were 24 females (24.0%) and 76 males (76.0%). This indicates that males were more common in the CAD group than in the no-CAD group

2.2 c)

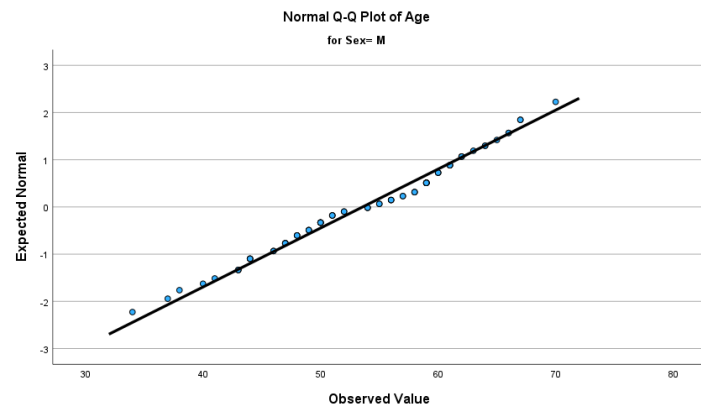
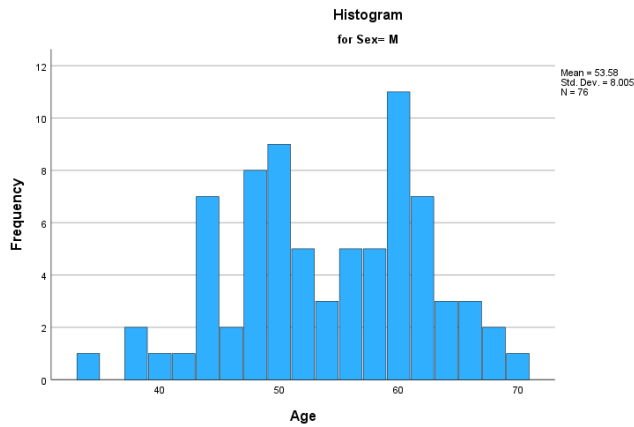
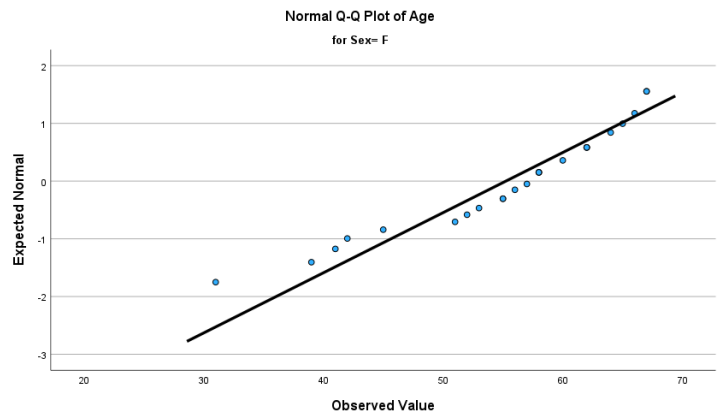
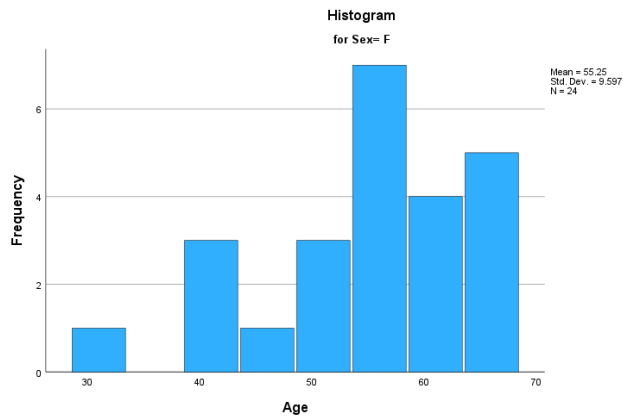
Although the decision was made using the 0.05 significance level, the probability of being wrong is given by the observed p-value. For age, $p < 0.001$, so the probability of being wrong is less than 0.1%.

2.3)

Assessing the normality of the age distributions among women and men with CAD.

Tests of Normality							
Age	Sex	Kolmogorov-Smirnov ^a			Shapiro-Wilk		
		Statistic	df	Sig.	Statistic	df	Sig.
Age	F	.156	24	.134	.915	24	.046
	M	.106	76	.034	.980	76	.261

a. Lilliefors Significance Correction



H0: Age is normally distributed in women/men with CAD **H1:** Age is not normally distributed in women/ men with CAD

Normality of age was tested separately in female and male patients with CAD using the Shapiro–Wilk test. In female CAD patients, the p-value was 0.046, which was less than 0.05, the null hypothesis was rejected, indicating that age was not normally distributed. In male CAD patients, the p-value was 0.261, which was greater than 0.05, cannot reject the null hypothesis, indicating that age was approximately normally distributed. Since normality was not satisfied in both groups, a non-parametric test was used. Because female and male CAD patients are two independent groups, the Mann–Whitney U test was chosen to compare age between them.

A Mann–Whitney U test was used to compare age between women and men with CAD

Ranks				
	SEX_NUM	N	Mean Rank	Sum of Ranks
Age	F	24	56.35	1352.50
	M	76	48.65	3697.50
	Total	100		

Test Statistics	
	Age
Mann-Whitney U	771.500
Wilcoxon W	3697.500
Z	-1.135
Asymp. Sig. (2-tailed)	.256

a. Grouping Variable: SEX_NUM

H0: women / men with CAD have the same age distribution

H1: women / men with CAD have different age distributions

Among patients with CAD, a Mann–Whitney U test was used to compare age between women and men. The test showed no statistically significant difference in age distribution between the two groups ($U = 771.500$, $Z = -1.135$, $p = 0.256$). Since the p-value was greater than 0.05, the null hypothesis was not rejected. This means that the age distribution of women and men with CAD was not significantly different. Although the mean rank of age was slightly higher in women (56.35) than in men (48.65), this difference was not statistically significant.

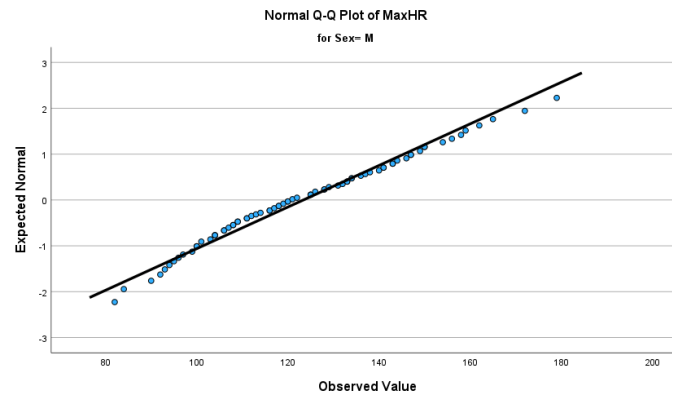
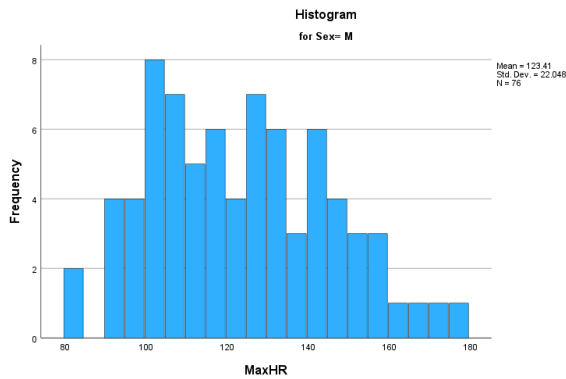
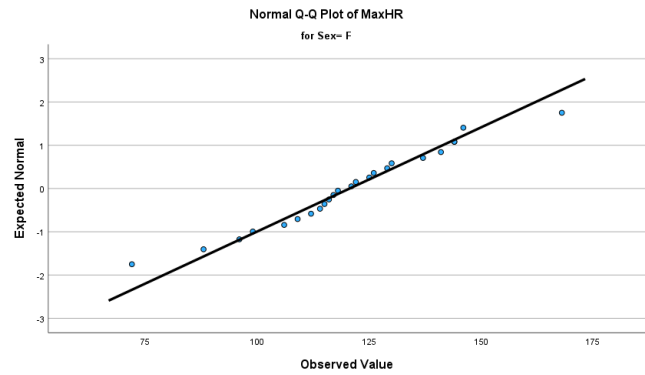
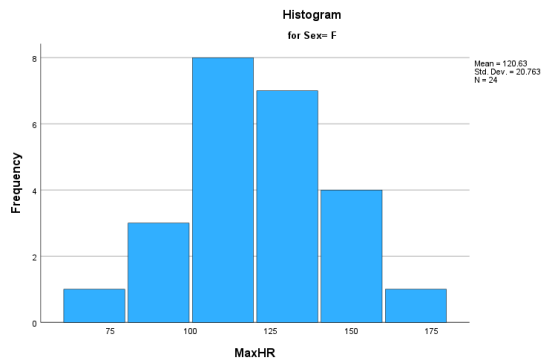
2.3 b)

Assessing the normality of the MaxHR distributions among women and men with CAD.

Tests of Normality							
		Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	Sex	Statistic	df	Sig.	Statistic	df	Sig.
MaxHR	F	.089	24	.200*	.985	24	.964
	M	.072	76	.200*	.980	76	.289

*. This is a lower bound of the true significance.

a. Lilliefors Significance Correction



H0: maxHR is normally distributed in women/ men with CAD

H1: maxHR is not normally distributed in women / men with CAD

Normality of maxHR was assessed separately in women and men with CAD using the Shapiro–Wilk test. Since both p-values were greater than 0.05, so null hypothesis cannot be rejected. So maxHR was considered approximately normally distributed in both groups. Therefore, an independent-samples t test was used to compare maxHR between women and men with CAD.

An independent-samples T test was used to compare maxHR between women and men with CAD

Group Statistics

	SEX_NUM	N	Mean	Std. Deviation	Std. Error Mean
MaxHR	F	24	120.63	20.763	4.238
	M	76	123.41	22.048	2.529

Independent Samples Test

			t-test for Equality of Means			
			t	df	Significance	
					One-Sided p	Two-Sided p
MaxHR	Equal variances assumed		-.546	98	.293	.586
	Equal variances not assumed		-.564	40.713	.288	.576

Independent Samples Test

t-test for Equality of Means				
			Std. Error Difference	95% Confidence Interval of the Difference Lower
		Mean Difference		
MaxHR	Equal variances assumed	-2.783	5.093	-12.891
	Equal variances not assumed	-2.783	4.936	-12.753

Independent Samples Test

		t-test for Equality of Means
		95% Confidence Interval of the Difference
		Upper
MaxHR	Equal variances assumed	7.325
	Equal variances not assumed	7.187

H0: Mean of MaxHR women = mean of MaxHR of men **H1:** mean of MaxHR of women $\neq$ mean of MaxHR of men

Among patients with CAD, an independent-samples t test was used to compare maxHR between women and men. The mean maxHR was 120.63 (SD = 20.76) in women and 123.41 (SD = 22.05) in men. The test showed no statistically significant difference between the two groups. Using the equal variances assumed row, the result was $t(98) = -0.546$, $p = 0.586$. Since the p-value was greater than 0.05, the null hypothesis was not rejected. This means that the mean maxHR did not differ significantly between women and men with CAD.

3) Correlation and regression between two variables

Since there variables are not normally distributed, Spearman correlation was used to test Correlation and regression between these variables

Patient with CAD (CAD=1)

Correlations			ST-DEP	ST/HR	RWA
Spearman's rho	ST-DEP	Correlation Coefficient	1.000	.783***	.078
		Sig. (2-tailed)	.	<.001	.440
		N	100	100	100
	ST/HR	Correlation Coefficient	.783***	1.000	.012
		Sig. (2-tailed)	<.001	.	.909
		N	100	100	100
	RWA	Correlation Coefficient	.078	.012	1.000
		Sig. (2-tailed)	.440	.909	.
		N	100	100	100

***. Correlation is significant at the 0.001 level (2-tailed).

H0: There is **no correlation** between the two variables.

H1: There **is a correlation** between the two variables.

In the CAD group, Spearman's rank correlation showed a strong positive statistically significant association between ST-DEP and ST/HR ($r_s=0.783$, $p<0.001$) Thus, the null hypothesis of no correlation was rejected for this pair. In contrast, no statistically significant correlations were found between ST-DEP and RWA ($r_s=0.078$, $p=0.440$) or between ST/HR and RWA ($r_s=0.012$, $p=0.909$) For these two pairs, the null hypothesis was not rejected.

Patient with no CAD (CAD=0)

Correlations			ST-DEP	ST/HR	RWA
Spearman's rho	ST-DEP	Correlation Coefficient	1.000	.724***	.099
		Sig. (2-tailed)	.	<.001	.326
		N	100	100	100
	ST/HR	Correlation Coefficient	.724***	1.000	.032
		Sig. (2-tailed)	<.001	.	.753
		N	100	100	100
	RWA	Correlation Coefficient	.099	.032	1.000
		Sig. (2-tailed)	.326	.753	.
		N	100	100	100

***. Correlation is significant at the 0.001 level (2-tailed).

H0: There is **no correlation** between the two variables.

H1: There is **a correlation** between the two variables

In the non-CAD group, Spearman's rank correlation showed a strong positive statistically significant association between ST-DEP and ST/HR ($r_s=0.724$, $p<0.001$) Thus, the null hypothesis of no correlation was rejected for this pair. In contrast, no statistically significant correlations were found between ST-DEP and RWA ($r_s=0.099$, $p=0.326$) or between ST/HR and RWA ($r_s=0.032$, $p=0.753$). Therefore, the null hypothesis was not rejected for these pairs. Overall, similar findings were observed in both CAD and non-CAD groups: only ST-DEP and ST/HR showed a significant correlation, whereas RWA was not significantly correlated with either variable.

3.2)

simple linear regression **Patient with no CAD (CAD=0)**

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.506 ^a	.256	.248	18.384

a. Predictors: (Constant), Age

b. Dependent Variable: MaxHR

ANOVA

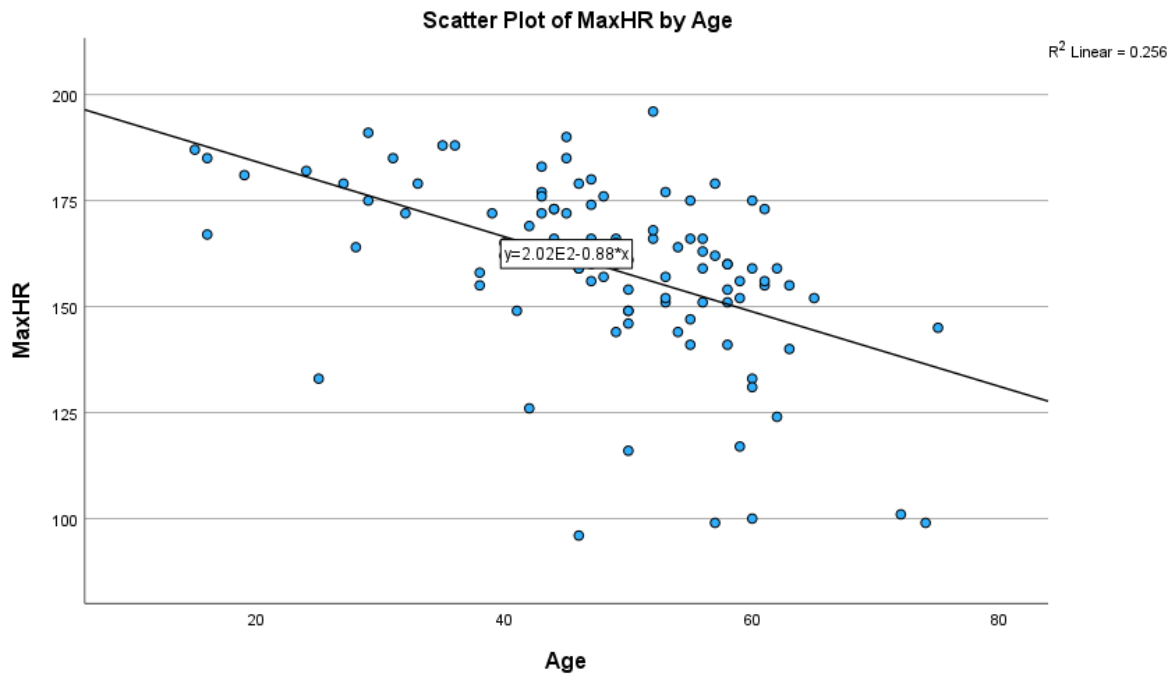
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	11389.179	1	11389.179	33.699	<.001 ^b
	Residual	33121.331	98	337.973		
	Total	44510.510	99			

a. Dependent Variable: MaxHR

b. Predictors: (Constant), Age

Coefficients

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	95.0% Confidence Interval for B	
		B	Std. Error	Beta			Lower Bound	Upper Bound
1	(Constant)	201.821	7.591		26.589	<.001	186.758	216.885
	Age	-.884	.152	-.506	-5.805	<.001	-1.186	-.582



H0: There is no linear relationship between age and maxHR.

H1: There is a linear relationship between age and maxHR.

In patients without CAD, simple linear regression was used to examine the relationship between age and maximum heart rate (maxHR). The regression model was statistically significant ($F(1,98)=33.699$, $p<0.001$), so the null hypothesis of no linear relationship was rejected.

The regression equation is **maxHR = 201.821 – 0.884 × age**

This shows that maxHR decreases as age increases.

The 95% confidence interval for the intercept was **[186.758, 216.885]**, while the 95% confidence interval for the slope was **[-1.186, -0.582]**. Therefore, we are 95% confident that the true intercept and slope of the regression line fall within these intervals.

simple linear regression **Patient with CAD (CAD=1)**

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.341 ^a	.116	.107	20.480

a. Predictors: (Constant), Age

b. Dependent Variable: MaxHR

ANOVA

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	5412.176	1	5412.176	12.904	<.001 ^b
	Residual	41103.064	98	419.419		
	Total	46515.240	99			

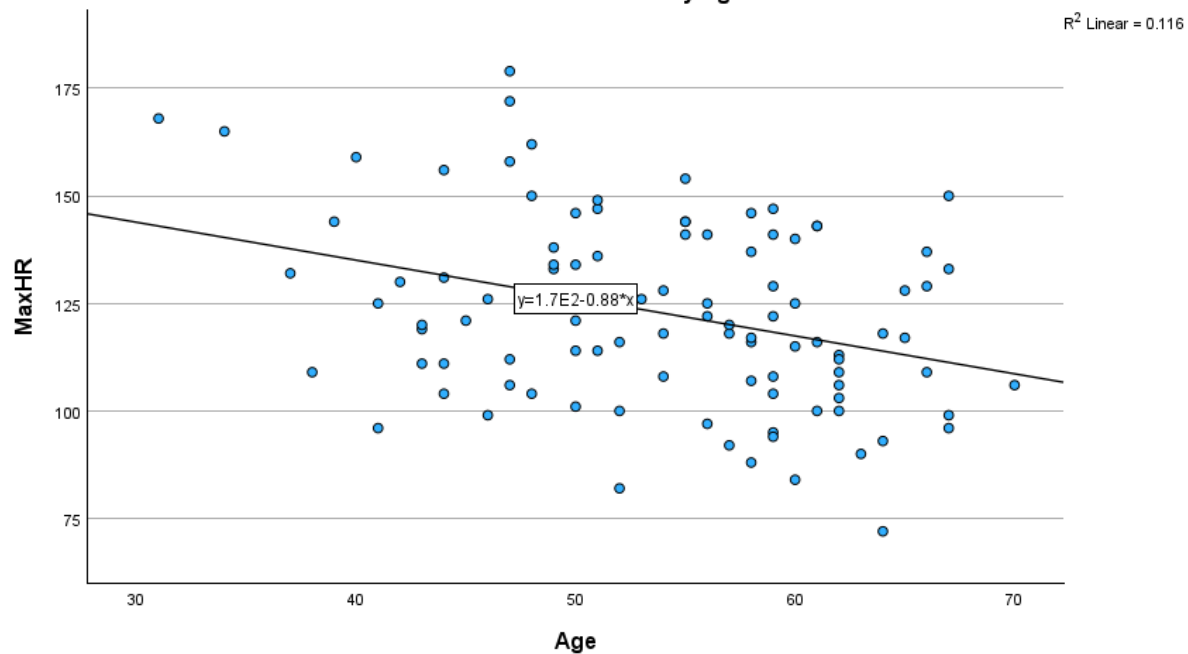
a. Dependent Variable: MaxHR

b. Predictors: (Constant), Age

Coefficients

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	95.0% Confidence Interval for B	
		B	Std. Error	Beta			Lower Bound	Upper Bound
1	(Constant)	170.286	13.393		12.714	<.001	143.707	196.865
	Age	-.881	.245	-.341	-3.592	<.001	-1.367	-.394

Scatter Plot of MaxHR by Age



H0: There is no linear relationship between age and maxHR.

H1: There is a linear relationship between age and maxHR.

In patients with CAD, simple linear regression was used to examine the relationship between age and maximum heart rate (maxHR). The regression model was statistically significant [F(1,98)=12.904, p<0.001)]. So the null hypothesis was rejected. The regression equation was

$$\text{maxHR} = 170.286 - 0.881 \times \text{age}$$

Indicating that maxHR decreases with increasing age. The 95% confidence interval for the intercept was [143.707, 196.865], and the 95% confidence interval for the slope was [-1.367, -0.394]. Thus, we are 95% confident that the true intercept and slope lie within these intervals.

3.3)

Simple linear regression was used to examine the relationship between maxHR and ST-DEP

Model Summary^b

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.294 ^a	.086	.082	.09576

a. Predictors: (Constant), MaxHR

b. Dependent Variable: ST-DEP

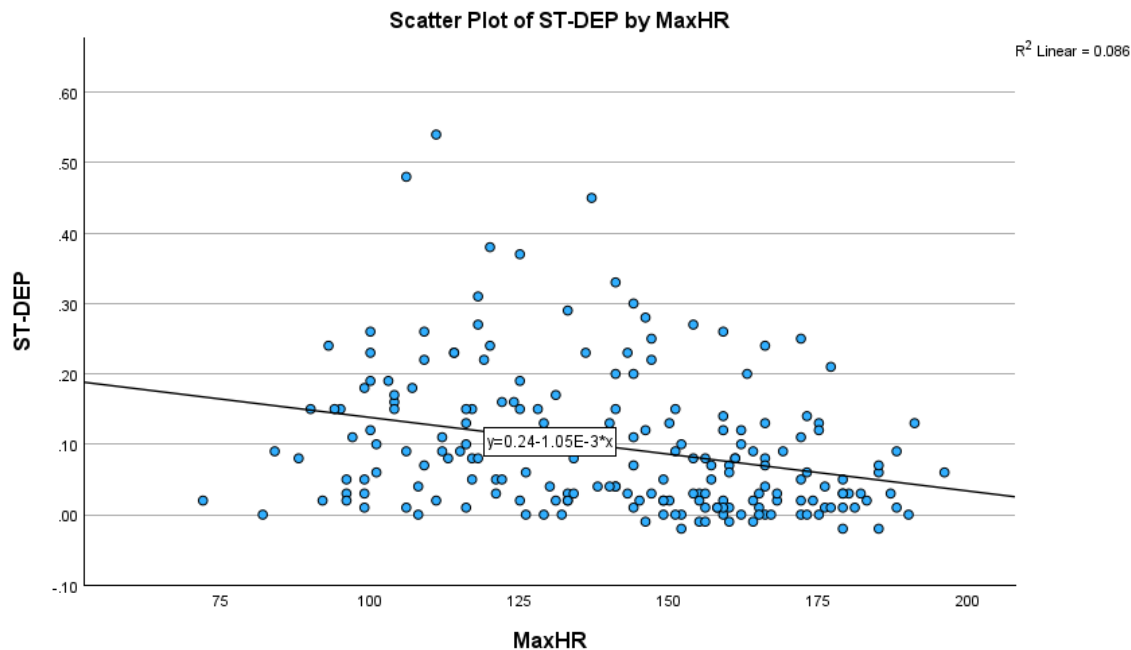
ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	.172	1	.172	18.707	<.001 ^b
	Residual	1.816	198	.009		
	Total	1.987	199			

a. Dependent Variable: ST-DEP

b. Predictors: (Constant), MaxHR

		Coefficients ^a					95.0% Confidence Interval for B	
		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Lower Bound	Upper Bound
Model		B	Std. Error	Beta				
1	(Constant)	.243	.035		6.987	<.001	.174	.311
	MaxHR	-.001	.000	-.294	-4.325	<.001	-.002	-.001



H₀: There is no linear relationship between maxHR and ST depression.

H₁: There is a linear relationship between maxHR and ST depression.

Simple linear regression was used to examine the relationship between maximum heart rate (maxHR) and ST depression (ST-DEP) in the whole sample. The regression model was statistically significant ($F(1,198)=18.707$, $p<0.001$). So the null hypothesis was rejected. The regression equation was

$$\text{ST-DEP} = 0.243 - 0.001 \times \text{maxHR}$$

Indicating that ST depression decreases as maxHR increases. The 95% confidence interval for the intercept was [0.174, 0.311], and the 95% confidence interval for the slope was [-0.002, -0.001]. Thus, we are 95% confident that the true intercept and slope lie within these intervals.

Simple linear regression was used to examine the relationship between maxHR and ST/HR

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.552 ^a	.304	.301	1.84674

a. Predictors: (Constant), MaxHR

b. Dependent Variable: ST/HR

ANOVA

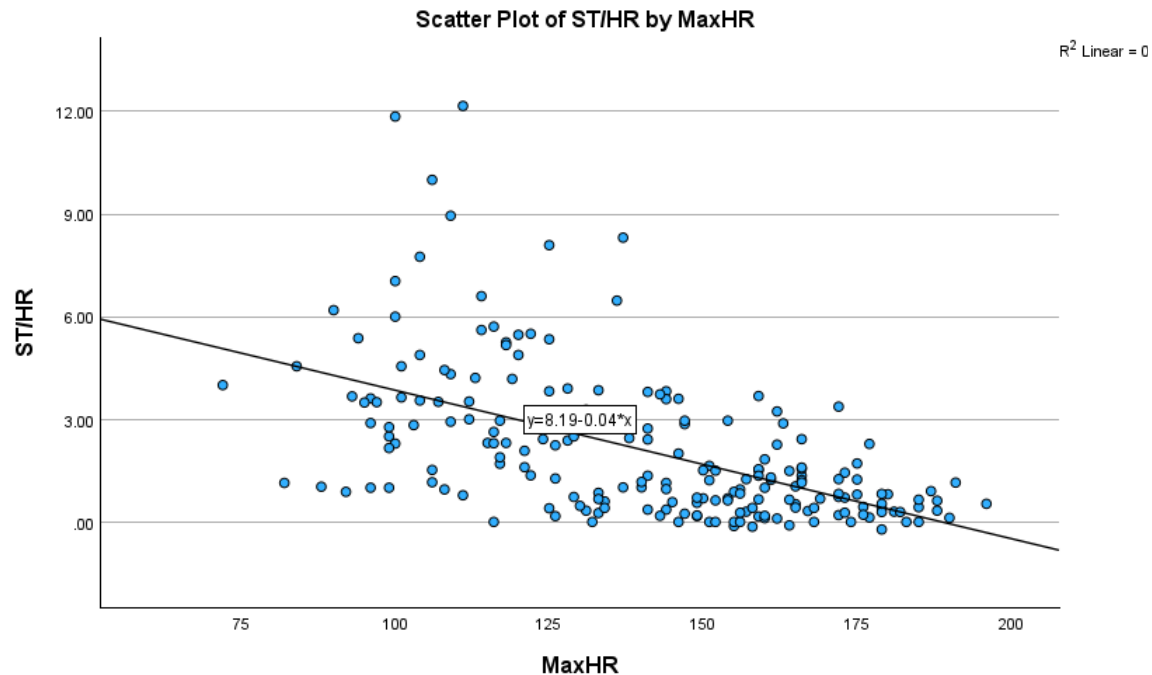
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	295.412	1	295.412	86.619	<.001 ^b
	Residual	675.272	198	3.410		
	Total	970.683	199			

a. Dependent Variable: ST/HR

b. Predictors: (Constant), MaxHR

Coefficients

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	95.0% Confidence Interval for B	
		B	Std. Error	Beta			Lower Bound	Upper Bound
1	(Constant)	8.188	.670		12.229	<.001	6.867	9.508
	MaxHR	-.043	.005	-.552	-9.307	<.001	-.053	-.034



H₀: There is no linear relationship between maxHR and ST/HR index.

H₁: There is a linear relationship between maxHR and ST/HR index.

Simple linear regression was used to examine the relationship between maximum heart rate (maxHR) and ST/HR index in the whole sample. The regression model was statistically significant ($F(1,198)=86.619, p<0.001$). So the null hypothesis was rejected. The regression equation was

$$\text{ST/HR} = 8.188 - 0.043 \times \text{maxHR}$$

indicating that the ST/HR index decreases as maxHR increases. The 95% confidence interval for the intercept was [6.867, 9.508], and the 95% confidence interval for the slope was [-0.053, -0.034]. Thus, we are 95% confident that the true intercept and slope lie within these intervals.

3.4)

As distributions are not normal, Spearman's rank correlation is used to calculate Correlations of age and RWA

Correlations			RWA	Age
Spearman's rho	RWA	Correlation Coefficient	1.000	.193**
		Sig. (2-tailed)	.	.006
		N	200	200
	Age	Correlation Coefficient	.193**	1.000
		Sig. (2-tailed)	.006	.
		N	200	200

**. Correlation is significant at the 0.01 level (2-tailed).

H₀: There is **no statistically significant correlation** between **Age** and **ΔRWA** in the total population.

H₁: There **is** a statistically significant correlation between **Age** and **ΔRWA** in the total population.

The age and ΔRWA variables were also tested in the total population using Spearman's rank correlation. A weak but statistically significant positive correlation was found between age and ΔRWA ($r_s = 0.193, p = 0.006, N = 200$)

Since p value (0.006) is less than 0.05, we can reject null hypothesis. So There is a statistically significant correlation between Age and ΔRWA in the total population. Although no significant association was found when CAD and non-CAD patients were analyzed separately, a significant correlation appeared in the pooled sample. This may be explained by differences between the two groups and by the larger total sample size, which increased the power to detect a weak overall association.

4. Estimating the accuracy of diagnostic classifiers

4.1) Sensitivity, specificity and diagnostic accuracy in the total population

CAD * STDEP_pos Crosstabulation

			STDEP_pos		Total
			0	1	
CAD	0	Count	81	19	100
		% within CAD	81.0%	19.0%	100.0%
		% within STDEP_pos	64.8%	25.3%	50.0%
	1	Count	44	56	100
		% within CAD	44.0%	56.0%	100.0%
		% within STDEP_pos	35.2%	74.7%	50.0%
Total	Count		125	75	200
	% within CAD		62.5%	37.5%	100.0%
	% within STDEP_pos		100.0%	100.0%	100.0%

1= ST depression ≥ 0.10

0= ST depression < 0.10

CAD * STHR_pos Crosstabulation

			STHR_pos		Total
			0	1	
CAD	0	Count	84	16	100
		% within CAD	84.0%	16.0%	100.0%
		% within STHR_pos	72.4%	19.0%	50.0%
	1	Count	32	68	100
		% within CAD	32.0%	68.0%	100.0%
		% within STHR_pos	27.6%	81.0%	50.0%
Total	Count		116	84	200
	% within CAD		58.0%	42.0%	100.0%
	% within STHR_pos		100.0%	100.0%	100.0%

1= ST/HR index $\geq 1.60\mu\text{V}/\text{beats}/\text{min}$

0= ST/HR index $< 1.60\mu\text{V}/\text{beats}/\text{min}$

CAD * RWA_pos Crosstabulation

			RWA_pos		Total
			0	1	
CAD	0	Count	9	91	100
		% within CAD	9.0%	91.0%	100.0%
		% within RWA_pos	52.9%	49.7%	50.0%
	1	Count	8	92	100
		% within CAD	8.0%	92.0%	100.0%
		% within RWA_pos	47.1%	50.3%	50.0%
Total	Count		17	183	200
	% within CAD		8.5%	91.5%	100.0%
	% within RWA_pos		100.0%	100.0%	100.0%

1 = $\Delta\text{RWA} \geq 0.0 \text{ mV}$

0= $\Delta\text{RWA} < 0.0 \text{ mV}$

Classifier	TP	FP	TN	FN	Sensitivity	Specificity	Diagnostic accuracy
ST-DEP ≥ 0.10	56	19	81	44	56.0%	81.0%	68.5%
ST/HR ≥ 1.60	68	16	84	32	68.0%	84.0%	76.0%
$\Delta RWA \geq 0.0$	92	91	9	8	92.0%	9.0%	50.5%

TP = The test says **positive**, and the patient **really has CAD**.

FP = The test says **positive**, but the patient **does not have CAD**.

TN = The test says **negative**, and the patient **does not have CAD**.

FN = The test says **negative**, but the patient **really has CAD**.

In the total population, **ST/HR ≥ 1.60** had the best diagnostic performance, with the highest accuracy (76.0%) and good balance between sensitivity (68.0%) and specificity (84.0%). **ST-DEP ≥ 0.10** showed moderate performance, while **$\Delta RWA \geq 0.0$** had very high sensitivity (92.0%) but extremely low specificity (9.0%), making it a poor classifier overall.

Sensitivity, specificity and diagnostic accuracy in the female population

CAD * STDEP_pos Crosstabulation

			STDEP_pos		
			0	1	Total
CAD	0	Count	36	14	50
		% within CAD	72.0%	28.0%	100.0%
		% within STDEP_pos	76.6%	51.9%	67.6%
	1	Count	11	13	24
		% within CAD	45.8%	54.2%	100.0%
		% within STDEP_pos	23.4%	48.1%	32.4%
Total	Count	47	27	74	
	% within CAD	63.5%	36.5%	100.0%	
	% within STDEP_pos	100.0%	100.0%	100.0%	

1= ST depression ≥ 0.10

0= ST depression < 0.10

CAD * STHR_pos Crosstabulation

		STHR_pos		Total
		0	1	
CAD	0	Count	38	12
		% within CAD	76.0%	24.0%
		% within STHR_pos	86.4%	40.0%
	1	Count	6	18
		% within CAD	25.0%	75.0%
		% within STHR_pos	13.6%	60.0%
Total	Count		44	30
	% within CAD		59.5%	40.5%
	% within STHR_pos		100.0%	100.0%

1= ST/HR index $\geq 1.60\mu\text{V}/\text{beats}/\text{min}$

0= ST/HR index $< 1.60\mu\text{V}/\text{beats}/\text{min}$

CAD * RWA_pos Crosstabulation

		RWA_pos		Total
		0	1	
CAD	0	Count	7	43
		% within CAD	14.0%	86.0%
		% within RWA_pos	58.3%	69.4%
	1	Count	5	19
		% within CAD	20.8%	79.2%
		% within RWA_pos	41.7%	30.6%
Total	Count		12	62
	% within CAD		16.2%	83.8%
	% within RWA_pos		100.0%	100.0%

1 = $\Delta\text{RWA} \geq 0.0 \text{ mV}$

0= $\Delta\text{RWA} < 0.0 \text{ mV}$

Classifier	TP	FP	TN	FN	Sensitivity	Specificity	Diagnostic accuracy
ST-DEP ≥ 0.10	13	14	36	11	54.2%	72.0%	66.2%
ST/HR ≥ 1.60	18	12	38	6	75.0%	76.0%	75.7%
$\Delta\text{RWA} \geq 0.0$	19	43	7	5	79.2%	14.0%	35.1%

TP = The test says **positive**, and the Females really **has** CAD.

FP = The test says **positive**, but the Females **does not have** CAD.

TN = The test says **negative**, and the Females **does not have** CAD.

FN = The test says **negative**, but the Females **really has** CAD.

In the female population, **ST/HR ≥ 1.60** showed the best diagnostic performance, with the highest diagnostic accuracy (75.7%) and a good balance between sensitivity (75.0%) and specificity (76.0%). **ST-DEP ≥ 0.10** showed moderate performance, while **$\Delta\text{RWA} \geq 0.0$** had high sensitivity (79.2%) but very low specificity (14.0%), making it a poor classifier overall.

Predictive values of ST/HR index ≥ 1.60 in the total and female populations

Population	TP	FP	TN	FN	PV+	PV–
Total population	68	16	84	32	81.0%	72.4%
Female population	18	12	38	6	60.0%	86.4%

TP: patient has CAD and the test is positive

FP: patient does not have CAD but the test is positive

TN: patient does not have CAD and the test is negative

FN: patient has CAD but the test is negative

PV+: proportion of test-positive patients who truly have CAD

PV–: proportion of test-negative patients who truly do not have CAD

In the total population, a positive ST/HR test (≥ 1.60) was quite reliable for indicating CAD, since PV+ was 81.0%. In the female population, PV+ was lower (60.0%), meaning that a positive test was less reliable. However, PV– was higher in women (86.4%), indicating that a negative test was more useful for excluding CAD in the female population.

4.2 a)

Sensitivity and specificity of ST/HR index at different partition values

CAD * STHR08 Crosstabulation

			STHR08		
			0	1	Total
CAD	0	Count	52	48	100
		% within CAD	52.0%	48.0%	100.0%
		% within STHR08	74.3%	36.9%	50.0%
	1	Count	18	82	100
		% within CAD	18.0%	82.0%	100.0%
		% within STHR08	25.7%	63.1%	50.0%
Total	Count	70	130	200	
	% within CAD	35.0%	65.0%	100.0%	
	% within STHR08	100.0%	100.0%	100.0%	

CAD * STHR12 Crosstabulation

			STHR12		
			0	1	Total
CAD	0	Count	67	33	100
		% within CAD	67.0%	33.0%	100.0%
		% within STHR12	70.5%	31.4%	50.0%
	1	Count	28	72	100
		% within CAD	28.0%	72.0%	100.0%
		% within STHR12	29.5%	68.6%	50.0%
Total	Count	95	105	200	
	% within CAD	47.5%	52.5%	100.0%	
	% within STHR12	100.0%	100.0%	100.0%	

CAD * STHR16 Crosstabulation

			STHR16		
			0	1	Total
CAD	0	Count	84	16	100
		% within CAD	84.0%	16.0%	100.0%
		% within STHR16	72.4%	19.0%	50.0%
	1	Count	32	68	100
		% within CAD	32.0%	68.0%	100.0%
		% within STHR16	27.6%	81.0%	50.0%
Total	Count	116	84	200	
	% within CAD	58.0%	42.0%	100.0%	
	% within STHR16	100.0%	100.0%	100.0%	

CAD * STHR20 Crosstabulation

			STHR20		
			0	1	Total
CAD	0	Count	88	12	100
		% within CAD	88.0%	12.0%	100.0%
		% within STHR20	72.1%	15.4%	50.0%
	1	Count	34	66	100
		% within CAD	34.0%	66.0%	100.0%
		% within STHR20	27.9%	84.6%	50.0%
Total	Count	122	78	200	
	% within CAD	61.0%	39.0%	100.0%	
	% within STHR20	100.0%	100.0%	100.0%	

CAD * STHR24 Crosstabulation

			STHR24		
			0	1	Total
CAD	0	Count	92	8	100
		% within CAD	92.0%	8.0%	100.0%
		% within STHR24	69.2%	11.9%	50.0%
	1	Count	41	59	100
		% within CAD	41.0%	59.0%	100.0%
		% within STHR24	30.8%	88.1%	50.0%
Total	Count	133	67	200	
	% within CAD	66.5%	33.5%	100.0%	
	% within STHR24	100.0%	100.0%	100.0%	

ST/HR cutoff ($\mu\text{V}/\text{bpm}$)	TP	FP	TN	FN	Sensitivity	Specificity
0.8	82	48	52	18	82.0%	52.0%
1.2	72	33	67	28	72.0%	67.0%
1.6	68	16	84	32	68.0%	84.0%
2.0	66	12	88	34	66.0%	88.0%
2.4	59	8	92	41	59.0%	92.0%

TP: CAD present and test positive

FP: CAD absent but test positive

TN: CAD absent and test negative

FN: CAD present but test negative

STHR08 = ST/HR index classified using cutoff **0.8 $\mu\text{V}/\text{bpm}$**
positive if **ST/HR $\geq$ 0.8**

STHR12 = ST/HR index classified using cutoff **1.2 $\mu\text{V}/\text{bpm}$**
positive if **ST/HR $\geq$ 1.2**

STHR16 = ST/HR index classified using cutoff **1.6 $\mu\text{V}/\text{bpm}$**
positive if **ST/HR $\geq$ 1.6**

STHR20 = ST/HR index classified using cutoff **2.0 $\mu\text{V}/\text{bpm}$**
positive if **ST/HR $\geq$ 2.0**

STHR24 = ST/HR index classified using cutoff **2.4 $\mu\text{V}/\text{bpm}$**
positive if **ST/HR $\geq$ 2.4**

The table shows the expected trade-off between sensitivity and specificity. When the ST/HR cutoff increased, sensitivity became lower and specificity became higher. Thus, lower cutoffs were better for detecting CAD, while higher cutoffs were better for excluding patients without CAD. Among these values, **2.0 $\mu\text{V}/\text{bpm}$** provided a good balance between the two.

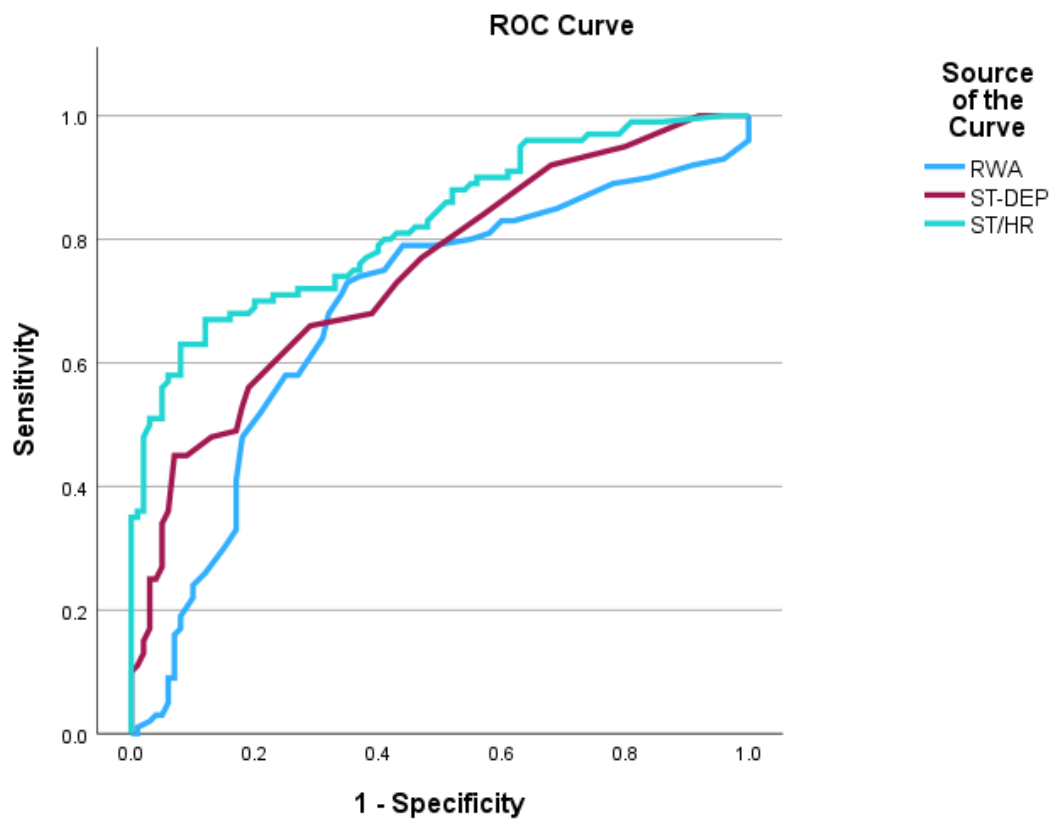
4.2 b)

ST/HR cutoff ($\mu\text{V}/\text{bpm}$)	Sensitivity	Specificity	Accuracy at 20% prevalence
0.8	82.0%	52.0%	58.0%
1.2	72.0%	67.0%	68.0%
1.6	68.0%	84.0%	80.8%
2.0	66.0%	88.0%	83.6%
2.4	59.0%	92.0%	85.4%

$$\text{Accuracy} = 0.20 \times \text{Sensitivity} + 0.80 \times \text{Specificity}$$

The highest diagnostic accuracy was obtained with the cutoff **2.4 $\mu\text{V}/\text{bpm}$** (85.4%). Therefore, when the prevalence of CAD is 20%, the best partition value is **2.4 $\mu\text{V}/\text{bpm}$** .

4.2 c)



Area Under the ROC Curve

Test Result Variable(s)	Area
ST-DEP	.747
ST/HR	.824
RWA	.683

The test result variable(s): ST-DEP, ST/HR, RWA has at least one tie between the positive actual state group and the negative actual state group. Statistics may be biased.

ROC curves were constructed for ST depression, ST/HR index, and Δ RWA, and all curves were presented in the same figure. The area under the ROC curve was used to compare the overall diagnostic performance of the classifiers without selecting a single partition value. The AUC values were 0.747 for ST depression, 0.824 for ST/HR index, and 0.683 for Δ RWA. Thus, the ST/HR index had the largest AUC and therefore the best overall diagnostic performance, while Δ RWA had the weakest performance.

4.2 d)

Among the three diagnostic classifiers, the **ST/HR index** had the largest area under the ROC curve (AUC=0.824). Therefore, it showed the best overall ability to discriminate between patients with CAD and patients without CAD. ST depression had moderate performance (AUC=0.747), whereas Δ RWA had the smallest AUC (0.683), indicating the weakest diagnostic performance.

4.3 a)

Coordinates of the ROC Curve

Test Result Variable(s): ST-DEP

Positive if Greater Than or Equal To ^a	Sensitivity	1 - Specificity
-1.0200	1.000	1.000
-.0150	1.000	.970
-.0050	1.000	.920
.0050	.950	.800
.0150	.920	.680
.0250	.840	.570
.0350	.770	.470
.0450	.730	.430
.0550	.680	.390
.0650	.670	.340
.0750	.660	.290
.0850	.610	.240
.0950	.560	.190
.1050	.530	.180
.1150	.490	.170
.1250	.480	.130
.1350	.450	.090
.1450	.450	.070
.1550	.360	.060
.1650	.340	.050
.1750	.320	.050
.1850	.300	.050
.1950	.270	.050
.2050	.250	.040
.2150	.250	.030
.2250	.220	.030
.2350	.170	.030
.2450	.150	.020
.2550	.130	.020
.2650	.110	.010
.2750	.100	.000
.2850	.090	.000
.2950	.080	.000
.3050	.070	.000
.3200	.060	.000
.3500	.050	.000
.3750	.040	.000
.4150	.030	.000
.4650	.020	.000
.5100	.010	.000
1.5400	.000	.000

The test result variable(s): ST-DEP has at least one tie between the positive actual state group and the negative actual state group.

a. The smallest cutoff value is the minimum observed test value minus 1, and the largest cutoff value is the maximum observed test value plus 1. All the other cutoff values are the averages of two consecutive ordered observed test values.

Coordinates of the ROC Curve

Test Result Variable(s): ST/HR

Positive if Greater Than or Equal To ^a	Sensitivity	1 - Specificity
-1.2200	1.000	1.000
-.1800	1.000	.990
-.1300	1.000	.980
-.1100	1.000	.970
-.0500	1.000	.960
.0500	.990	.860
.1050	.990	.850
.1150	.990	.840
.1250	.990	.830
.1450	.990	.820
.1650	.990	.810
.1750	.980	.800
.1850	.970	.790
.1950	.970	.780
.2050	.970	.770
.2250	.970	.760
.2500	.970	.750
.2650	.970	.740
.2800	.960	.730
.2950	.960	.710
.3100	.960	.690
.3250	.960	.680
.3450	.960	.660
.3800	.960	.640
.4050	.950	.630
.4150	.920	.630
.4250	.910	.630
.4350	.910	.620
.4550	.910	.610
.4950	.900	.610
.5250	.900	.600
.5450	.900	.580
.5700	.900	.570
.5900	.900	.560
.6100	.890	.560
.6250	.890	.550
.6400	.880	.540
.6550	.880	.520
.6650	.870	.520
.6750	.860	.520
.6850	.860	.510
.7000	.850	.500
.7200	.840	.490
.7550	.830	.480
.7900	.820	.480
.8050	.820	.470
.8150	.820	.460
.8250	.810	.450
.8400	.810	.440
.8650	.810	.430
.8900	.800	.420
.9250	.800	.410
.9550	.790	.400
.9800	.780	.400
1.0050	.770	.380
1.0200	.760	.370
1.0350	.750	.370
1.0900	.750	.360
1.1450	.740	.350
1.1550	.740	.330
1.1650	.730	.330
1.1950	.720	.330
1.2250	.720	.310
1.2350	.720	.300
1.2450	.720	.290
1.2600	.720	.270
1.2850	.710	.270
1.3250	.710	.260
1.3550	.710	.230
1.4000	.700	.230
1.4650	.700	.220
1.4950	.700	.210
1.5050	.700	.200
1.5150	.690	.200
1.5300	.680	.190
1.5450	.680	.180
1.5700	.680	.170
1.5950	.680	.160
1.6200	.670	.160
1.6700	.670	.150
1.7050	.670	.140
1.7700	.670	.130
1.8600	.670	.120

Coordinates of the ROC Curve

Test Result Variable(s): RWA

Positive if Greater Than or Equal To ^a	Sensitivity	1 - Specificity
-1.0700	1.000	1.000
-.0650	.990	1.000
-.0550	.980	1.000
-.0400	.970	1.000
-.0250	.960	1.000
-.0150	.930	.960
-.0050	.920	.910
.0050	.900	.840
.0150	.890	.780
.0250	.850	.690
.0350	.830	.620
.0450	.830	.600
.0550	.810	.580
.0650	.800	.550
.0750	.790	.490
.0850	.790	.440
.0950	.750	.410
.1050	.740	.370
.1150	.730	.350
.1250	.710	.340
.1350	.680	.320
.1450	.640	.310
.1550	.580	.270
.1650	.580	.250
.1750	.550	.230
.1850	.520	.210
.1950	.480	.180
.2050	.410	.170
.2150	.350	.170
.2250	.330	.170
.2350	.300	.150
.2450	.260	.120
.2550	.240	.100
.2650	.230	.100
.2750	.220	.100
.2850	.190	.080
.3000	.170	.080
.3150	.160	.070
.3250	.150	.070
.3400	.140	.070
.3600	.120	.070
.3800	.110	.070
.4050	.100	.070
.4300	.090	.070
.4450	.090	.060
.4550	.080	.060
.4650	.060	.060
.4850	.050	.060
.5250	.030	.050
.5600	.030	.040
.5750	.020	.030
.5900	.010	.010
.6400	.000	.010

Diagnostic classifier	Cutoff at ~85% specificity	Specificity	Sensitivity
ST depression	~ 0.120	85%	~ 48.5%
ST/HR index	1.670	85%	67.0%
Δ RWA	0.235	85%	30.0%

Sensitivity values at 85% specificity were determined from the ROC curve coordinates. For the ST/HR index, the sensitivity at 85% specificity was **67.0%**. For Δ RWA, the sensitivity at 85% specificity was **30.0%**. For ST depression, no exact point at 85% specificity was available in the ROC coordinates, so the value was estimated from the nearest points, giving a sensitivity of approximately **48.5%**. Thus, at 85% specificity, the **ST/HR index** had the highest sensitivity among the three classifiers.

4.3 b)

McNemar's test was used because the sensitivity values were compared between two diagnostic methods applied to the same patients, resulting in paired binary outcomes.

H₀: There is no significant difference in sensitivity at 85% specificity between ST depression and ST/HR index.

H₁: There is a significant difference in sensitivity at 85% specificity between ST depression and ST/HR index.

STDEP85 * STHR85 Crosstabulation

			STHR85		
			0	1	Total
STDEP85	0	Count	32	20	52
		% within STDEP85	61.5%	38.5%	100.0%
		% within STHR85	97.0%	29.9%	52.0%
	1	Count	1	47	48
		% within STDEP85	2.1%	97.9%	100.0%
		% within STHR85	3.0%	70.1%	48.0%
Total	Count	33	67	100	
	% within STDEP85	33.0%	67.0%	100.0%	
	% within STHR85	100.0%	100.0%	100.0%	

Chi-Square Tests

	Value	Exact Sig. (2-sided)
McNemar Test		<.001 ^a
N of Valid Cases	100	

a. Binomial distribution used.

STDEP85 = ST depression test result at the cutoff chosen for about **85% specificity**

STHR85 = ST/HR index test result at the cutoff chosen for about **85% specificity**

0 = negative test result

1 = positive test result

To compare the sensitivity values at 85% specificity between ST depression and ST/HR index, McNemar's test was performed in patients with CAD only. The test showed a statistically significant difference between the two classifiers ($p < 0.001$). As P value is less than 0.05, we null hypothesis is rejected. So, there is a significant difference in sensitivity at 85% specificity between ST depression and ST/HR index. This indicates that the sensitivity of the ST/HR index at 85% specificity was significantly higher than that of ST depression.

5) Feedback

This course was moderately challenging, and I learned many new things from it. I gained a better understanding of how SPSS can be used to analyze data, which was very valuable for me. The project work took more time than I had originally planned, but it was also a good learning experience. Thank you.